AMERICAN PSYCHOLOGICAL ASSOCIATION

Psychology of Violence

ISSN: 2152-0828

2024, Vol. 14, No. 2, 77–86
https://doi.org/10.1037/vio0000502

Estimating the Association Between Spanking and Early Childhood Development Using Between- and Within-Child Analyses

Jorge Cuartas
Harvard Graduate School of Education, Harvard University
Centro de Estudios sobre Seguridad y Drogas, Universidad de los Andes

Objective: Most research establishing the association between spanking and child development has used cross-sectional data and correlational methods in high-income countries, raising questions about the internal (i.e., causality) and external (e.g., generalizability) validity of this work. The main objective of this study is to use more internally valid methods to assess the links between spanking and developmental outcomes in four low- and-middle-income countries. ***Method:*** This study employs data from four longitudinal studies conducted in Bhutan, Cambodia, Ethiopia, and Rwanda, comprising 3,048 young children ($M_{\text{age}} = 53.11$ months, $SD = 9.16$ in the first round of data collection), which used comparable information on parent reports of spanking and direct assessments of children's development measured by the International Development and Early Learning Assessment. ***Results:*** Findings from random- and fixed-effects models that use both between- and within-child variation show that spanking relates to an array of negative cognitive, social–emotional, and motor outcomes. Yet, the results also demonstrate heterogeneity in estimates between sites, which warrants further investigation. ***Conclusions:*** These results, along with prior evidence, indicate that spanking might harm the early developmental skills that serve as children's foundation for lifelong well-being. Therefore, considering issues of human rights, existing evidence, and following a precautionary principle, policymakers, and practitioners should invest in strategies aimed at preventing and reducing spanking and other forms of violence against children across the world.

Keywords: spanking, physical punishment, early childhood development, low- and- middle-income countries, within-child analysis

Supplemental materials: https://doi.org/10.1037/vio0000502.supp

Scientific advances in understanding the biology and psychology of early childhood development have shed light on how early experience shapes the architecture of the developing brain and foundational cognitive, motor, and social–emotional skills (Black et al., 2017). As early experiences shape brain and skill development, they also establish the foundations of lifelong development, learning, and health (Berens & Nelson, 2019). In particular, there is scientific agreement that parenting behaviors (e.g., interactions, stimulation, discipline) provide fundamental inputs for children to achieve their developmental potential (Black et al., 2017; McCoy et al., 2022). Yet, there is still debate on the exact impacts that these parenting behaviors have on children in different settings due to vast variability in what is considered valuable, normative, and effective for raising children across cultures (Cuartas, 2023; Gershoff et al., 2010; Mesman et al., 2018).

One parenting behavior that has been the subject of extensive academic and societal controversy is physical punishment, in particular spanking or openhanded hitting with the intention of modifying child behavior (Gershoff & Grogan-Kaylor, 2016). Even though the United Nations Committee on the Rights of the Child has stated that physical punishment is a form of violence and a violation of children's rights (United Nations Committee on the Rights of the Child, 2007), the spanking of young children remains highly prevalent and socially normative. Globally, about two out of every three children aged 2–4 years were spanked in the month preceding household surveys conducted in the years before the COVID-19 pandemic, and growing evidence indicates that physical punishment has increased since the outbreak of coronavirus (Bhatia et al., 2021). To date, there are still some conflicting arguments on the specific developmental consequences of spanking during early childhood (Cuartas, 2023; Heilmann et al., 2021; Larzelere et al., 2018). In the current article, I address several of these controversies by providing more internally valid evidence about the links between spanking and developmental outcomes in four low- and middle-income countries (LMICs).

Plausible Mechanisms Linking Spanking and Child Development

Multiple biological and psychological mechanisms might explain why spanking could impact young children's developmental outcomes. Traditional models of human development like social

This article was published Online First January 11, 2024.
Jorge Cuartas https://orcid.org/0000-0002-0451-3298
The author thanks Dana McCoy, Aisha Yousafzai, Charles Nelson, Hirokazu Yoshikawa, Frances Gardner, and the participants of The International Society for the Study of Behavioural Development conference in Rhodes for their feedback. He also thanks Juliana Sánchez for her help in processing the figures and Save the Children for granting him access to the data. This project was supported by the National Academy of Education/Spencer Dissertation Fellowship and a fellowship from the American Psychological Foundation.

All data used in this study are available upon request to Save the Children (https://idela-network.org/).
Correspondence concerning this article should be addressed to Jorge Cuartas, Harvard Graduate School of Education, Harvard University, 13 Appian Way, Cambridge, MA 02138, United States. Email: jcuartas@g.harvard.edu

learning theory (Bandura, 1986), social information processing theory (Dodge, 1993), and attachment theory (Bowlby, 1969) suggest that spanking can lead to negative developmental consequences, particularly in the social–emotional domain. Briefly, researchers have used these perspectives to hypothesize that caregivers' use of spanking could inculcate expectations of aggression, intentionally or unintentionally model aggression as a means to solve problems, and erode the attachment bond, triggering social–emotional difficulties like internalizing and externalizing behavior problems as well as impaired self-regulation and conflict-solving skills (Cuartas, 2023; Gershoff, 2002).

More recently, researchers have employed neurodevelopmental perspectives like the dimensional model of adversity to hypothesize neural mechanisms linking spanking and negative cognitive and social–emotional outcomes (Cuartas et al., 2021). Quantitative and qualitative evidence from multiple settings indicates that spanking associates with emotional and physiological arousal in children, including fear, pain, distress, and atypical cortisol release (Bugental et al., 2003; Dobbs & Duncan, 2004; Gershoff, 2002; Gonzáles et al., 2019). Such automatic responses are thought to be adaptive to help children identify and respond to potential threats in the environment. Yet, they also affect highly plastic neural circuits in areas like the prefrontal cortex and hippocampus that underlie the development of higher order cognition, emotion and behavior regulation, and salience processing (Cuartas et al., 2021). In line with these theoretical predictions, emerging evidence shows atypical brain function (Cuartas et al., 2021) in children who were exposed to physical punishment—and spanking specifically—early in life relative to children who were not exposed.

Controversies in the Literature

Several systematic reviews and meta-analyses support the theoretical argument that spanking is detrimental for young children, finding consistent associations between spanking and impaired cognitive and social–emotional (e.g., self-regulation) development (Gershoff, 2002; Gershoff & Grogan-Kaylor, 2016; Heilmann et al., 2021). At the same time, other meta-analyses (Larzelere et al., 2018) and reviews of the literature (Cuartas, 2023) have raised questions regarding the internal and external validity of existing studies on spanking. In particular, the majority of published findings come from cross-sectional samples and use methodological approaches that rely on between-child variability and conventional covariate adjustment, limiting internal validity. Furthermore, most studies on spanking employ samples from the United States and other high-income countries, limiting external validity.

Prior studies estimating the effects of spanking on children have done so by comparing the outcomes of children who have and have not been spanked (i.e., using between-child variation). This is problematic because these groups of children are likely different in multiple ways beyond their exposure to spanking. In particular, myriad biological (e.g., genes), psychological (e.g., child behavior or caregiver depression), and social factors (e.g., social norms) might simultaneously predict caregivers' use of spanking and children's developmental outcomes, leading researchers to falsely conclude that spanking—rather than these other associated factors—is what is causing poor development (i.e., confounding; Duncan et al., 2004). Although including covariates can help to minimize the effects of confounding, many of these factors are not readily measurable, making it impossible to fully control for all potential confounders. Therefore, researchers have increasingly suggested the use of models that compare the same children's outcomes at times in their lives when they are and are not being spanked (i.e., within-child comparisons or using within-child variation), which reduces concerns about biases caused by differences in time-fixed characteristics, like genetics, between children. Researchers have also suggested robustness checks assessing differences between methodological approaches (e.g., using between- vs. within-child variation), which can strengthen the identification of arguably causal effects by testing the extent to which coefficients are robust to different sources of variation (Berry & Willoughby, 2017; Larzelere et al., 2018).

Similarly, the insufficient examination of links between spanking and developmental outcomes in diverse settings, including LMICs (Cuartas, 2021; Pace et al., 2019), makes it difficult to assess whether spanking might be universally detrimental. Some scholars have argued that the effects of spanking might vary depending on how normative physical punishment is in a given context (Deater-Deckard & Dodge, 1997). This cultural normativeness perspective suggests that as physical punishment is more normative within a setting, both caregivers and children might perceive it as less threatening, more just, or more legitimate, potentially mitigating its harmful impacts on child outcomes. Some researchers have even posited that spanking will not lead to negative outcomes if children perceive spanking as "just" or "normative" (Baumrind, 1997).

Emerging evidence from studies using international data from the Multiple Indicators Cluster Surveys indicates that spanking relates to negative developmental outcomes across many LMICs, regardless of the extent to which spanking is normative within countries, contradicting the cultural normativeness perspective (Cuartas, 2021; Grogan-Kaylor et al., 2021; Pace et al., 2019). These results might indicate that there might be more "universal" mechanisms linking spanking and child outcomes, such as those posited by psychological and neurodevelopmental frameworks. However, existing cross-cultural studies—including those using the Multiple Indicators Cluster Surveys—have largely relied on cross-sectional data, parent-reported, coarse measures of children's behaviors and skills coarse measures of early development that hardly capture the multidimensionality of young children's development (Waldman et al., 2021). Therefore, there is a need for further investigation of the links between spanking and early childhood development using more internally valid methodological approaches, cross-national samples, and comprehensive measures that capture the dimensionality of early development.

Objective and Hypothesis

In response to these gaps in knowledge, the present study employs data from four longitudinal studies conducted in LMICs with comparable measures of young children's exposure to spanking and direct assessments of early development. The main objective of the study is to provide more internally valid evidence on the links between young children's exposure to spanking and cognitive (i.e., numeracy and literacy skills), social–emotional, and motor skills in settings that have been largely overlooked in developmental and psychological science (Draper et al., 2022). Based on the theoretical framework, I hypothesized: (a) spanking will predict higher risk for detrimental social–emotional and cognitive (e.g., numeracy and literacy) outcomes and (b) spanking will not be associated with motor outcomes. This analysis provides novel evidence on the potential developmental consequences of spanking in underrepresented settings, which has implications for policy and practice aimed at mitigating psychosocial threats to young children's development.

Method

Data

Data were taken from four studies sponsored by Save the Children, each with two rounds of data collection, which included questionnaires for children's main caregivers and direct assessments of children's development. The first study was the National Early Childhood Care and Development Evaluation Study (Pisani et al., 2017; Save the Children, 2015) in Bhutan. This was a study with national coverage aimed at assessing the effects of early childhood care and development programs in the country. The second study was First Read (Pisani et al., 2016) in Cambodia, which conducted an observational evaluation of changes in children's development after Save the Children developed and provided books, parenting sessions, and community supports aimed at promoting children's development in 29 villages in Kampong Cham, Kratie, and Prey Veng provinces. The third study was the quasi-experimental longitudinal study in the Tigray Region (Seiden et al., 2018) in Ethiopia, which was an observational study of children's development in 40 schools. The final study was Advancing the Right to Read (Iwamoto et al., 2019) in Rwanda, which was a quasi-experimental evaluation of a program in Gasabo, Ngororero, Nyabihu, and Kikukiro districts. None of these programs specifically targeted physical punishment. Table 1 presents a brief description of each study, including dates for data collection, and Supplemental Material A presents more details on the procedures of each study. Although these studies collectively represent a diverse range of contexts, they are convenience-based and not nationally representative within each site. The total sample comprised 3,048 children aged, on average, between 50.41 (Bhutan) and 58.79 (Cambodia) months old in the first round of data collection and with a gap between the first and second rounds of data collection that ranged from 8 months in Bhutan to approximately 2 years in Ethiopia.

Measures

The four studies used comparable direct assessment of young children's development using the International Development and Early Learning Assessment (IDELA). The IDELA includes 22 subtasks, each with multiple items, that are conducted in approximately 30 min, aimed at obtaining valid and holistic assessments of children ages 3.5–6.5 years in different cultural contexts, particularly in LMICs (Pisani et al., 2018). Local experts translated and adapted IDELA's wording and stimulus materials in each country to ensure its acceptability, relevance, and validity. Prior psychometric and content analyses (e.g., Halpin et al., 2019; Pisani et al., 2018; Wolf et al., 2017), including the samples from this study, have examined the construct validity, interrater reliability, and internal consistency of IDELA, as well as convergent validity and test–retest reliability in subsamples. Findings from these prior psychometric analyses indicate that the IDELA reliably captures four domains of early development, including emergent numeracy, emergent literacy, social–emotional, and motor skills,

Table 1
Description of the Data Sets

Variable	Bhutan	Cambodia	Ethiopia	Rwanda
Name of study	Early Childhood Care and Development (ECCD) Evaluation Study	Cambodia first read	Quasi-experimental Longitudinal Study	Advancing the Right to Read Program (ARRP)
Number of children	1,377	382	693	596
Sample	National sample of children from 120 care and education centers across nine districts and a group of children not attending care and education centers.	Children from 29 villages in Kampong, Cham, Kratie, and Prey Veng	Children from eight woredas in the Tigray region	Children attending 60 schools in four districts in Gasabo, Ngororero, Nyabihu, and Kicukiro
First round of data collection	2015 March	2016	2017	2018
Second round of data collection	2015 November	2018	2019	2019
Spanked in T1	80%	64%	85%	59%
Numeracy in T1	25% (17) [α = .75]	39% (19) [α = .76]	35% (18) [α = .66]	30% (13) [α = .66]
Literacy in T1	15% (14) [α = .66]	34% (18) [α = .73]	21% (16) [α = .69]	23% (13) [α = .55]
Social–emotional in T1	21% (17) [α = .73]	40% (24) [α = .76]	29% (20) [α = .69]	28% (18) [α = .68]
Motor in T1	24% (25) [α = .75]	50% (30) [α = .76]	38% (22) [α = .61]	41% (22) [α = .65]
Male	50%	48%	51%	51%
M (*SD*) age in months in T1	50.41 (8.52)	58.79 (11.03)	54.80 (9.62)	53.23 (6.51)
Caregiver has at least secondary education	53%	85%	1%	21%
Mean number (*SD*) of different types of books available in T1	2.14 (1.60) [α = .75]	1.44 (1.05) [α = .50]	1.69 (1.32) [α = .63]	1.12 (0.95) [α = .51]
Mean number (*SD*) of different types of toys available in T1	4.58 (2.00) [α = .64]	4.39 (1.85) [α = .55]	3.50 (2.26) [α = .75]	2.62 (1.49) [α = .54]
Mean stimulation (*SD*) in T1	6.59 (2.38) [α = .79]	5.24 (2.12) [α = .67]	5.72 (2.98) [α = .86]	4.04 (2.63) [α = .79]

Note. α = Cronbach's α; T1 = Time 1.

but also indicate potential noninvariance across selected LMICs (Halpin et al., 2019). This means that associations of spanking with the IDELA can be interpreted in each of these country data sets, but means of the IDELA, for example, cannot readily be compared across settings (Halpin et al., 2019).

Table 1 presents Cronbach's α for internal consistency for each IDELA domain by site in this sample. The numeracy domain includes skills related to size and length discrimination, number identification, sorting, one-to-one correspondence, and simple problem solving, among others (39 items, α = .75). The literacy domain comprises skills like expressive vocabulary, emergent writing, sound discrimination, and listening comprehension (38 items, α = .66). The social–emotional domain includes skills related to self-awareness, emotion regulation, empathy, and problem solving (14 items, α = .73.). The motor domain includes fine and gross motor development (nine items, α = .75). Following the developers' scoring procedure, I used the percentage of correct items in each domain as main outcomes.

The studies also included a question for caregivers on the use of spanking. Caregivers in the four sites were asked: "in the past week, did you or any other family member spank the child?" Using such information, I followed extensive prior research (Gershoff & Grogan-Kaylor, 2016) to characterize children's exposure to spanking with a binary indicator that equaled one if the child was spanked and zero otherwise.

The data also included demographic and contextual variables reported by children's main caregivers, which I used as covariates in the analyses. In particular, there was information on children's age in months and sex, caregivers' age and education (none, primary, secondary, higher, or nonformal), and household wealth as measured by the first component of polychoric principal component analysis of basic dwelling characteristics (e.g., access to electricity) and household assets (e.g., television), as computed in Cuartas et al. (2023). Furthermore, the data included information on children's exposure to other harsh discipline at home, in particular whether caregivers yelled at the child (yes vs. no), a summed index of the availability of books in the home, comprising five dichotomous yes/no indicators (e.g., storybooks, coloring books; possible range = 0–5, α = .75), a summed index of playthings/toys in the home, comprising nine yes/no dichotomous indicators (e.g., toys from a shop, homemade toys; possible range = 0–9, α = .64), and an index of caregivers' stimulation practices, measured as a sum index of mothers', fathers', and other adult caregivers' engagement in nine play/learning activities with the child (yes vs. no), including reading, telling stories, singing songs, going outside, playing, naming objects, teaching new things, teaching letters, and playing counting games (α = .79). Table 1 presents descriptive information, including Cronbach's α, for each relevant scale by site, and Table 2 presents the correlation between the key study variables (spanking and

Table 2
Descriptive Statistics and Correlations for the Pooled Sample

Variable	Descriptive statistic							
	N	*M*	*SD*	1	2	3	4	5
T1								
1. Spanking	3,022	0.75	0.43	—				
2. Numeracy	3,048	30.19	17.44	−.03	—			
3. Literacy	3,048	20.65	16.09	−.06***	.63***	—		
4. Social–emotional	3,048	26.73	19.91	−.04*	.55***	.61***	—	
5. Motor	3,048	33.86	26.17	−.04*	.61***	.69***	.57***	—
Child is female	3,019	0.50	0.50	−.00	−.03	.03	.02	−.01
Child age (months)	3,018	53.11	9.16	−.04†	.35***	.38***	.29***	.44***
Mom age: <25 years	2,975	0.13	0.33	.04*	−.12***	−.15***	−.12***	−.14***
Mom age: 25–35 years	2,975	0.63	0.48	.02	.01	.03†	.05**	.03
Mom age: >36 years	2,975	0.24	0.43	−.05**	.08***	.08***	.04*	.08***
Mom edu.: none	2,970	0.39	0.48	.06***	−.04*	.11***	−.08***	−.05*
Mom edu.: primary	2,970	0.21	0.41	−.07***	−.01	.01	.01	.04*
Mom edu.: secondary	2,970	0.23	0.42	−.03	.10***	.21***	.17***	.13***
Mom edu.: higher	2,970	0.07	0.25	−.03	.08***	.09***	.04*	.04†
Mom edu.: nonformal	2,970	0.10	0.29	.06***	−.12***	−.20***	−.15***	−.20***
Wealth	2,987	−0.02	1.34	−.02	.14***	.21***	.16***	.18***
Books	2,957	1.75	1.42	.08***	.09***	.13***	.08***	.08***
Toys	2,970	3.95	2.10	.06***	.11***	.14***	.11***	.06***
Stimulation	2,978	5.73	2.71	.20***	.08***	.07***	.09***	.02
Yelling	3,019	0.58	0.49	.23***	.10***	.11***	.11***	.12***
T2								
1. Spanking	2,531	0.72	0.45	—				
2. Numeracy	2,623	55.70	23.09	−.15***	—			
3. Literacy	2,623	47.26	25.54	−.21***	.77***	—		
4. Social–emotional	2,623	51.19	24.98	−.11***	.66***	.62***	—	
5. Motor	2,623	66.17	26.91	−.19***	.66***	.73***	.54***	—
Books	2,508	2.20	1.56	.09***	.09***	.08***	.05*	.05**
Toys	2,492	5.03	2.29	.01	.15***	.13***	.10***	.11***
Stimulation	2,494	6.59	2.61	.29***	.02	−.03	.01	−.04*
Yelling	2,534	0.56	0.49	.23***	.17***	.08***	.15***	.05**

Note. edu. = education; T1 = Time 1; T2 = Time 2.
† $p < .1$. * $p < .05$. ** $p < .01$. *** $p < .001$.

developmental outcomes) and the demographic and contextual covariates.

Analytic Approach

Using these data, I conducted pooled and site-specific analyses on the links between children's exposure to spanking and each developmental domain, leveraging both between- and within-child variation. I included site fixed-effects (FE) in all pooled analyses to restrict comparisons to children living in the same site, considering the potential noninvariance of IDELA across LMICs (Halpin et al., 2019).

To begin, I follow standard procedures in the literature on spanking and used random-effects models that use between-child variation (i.e., comparisons between children), estimating a set of linear models with the following specification:

$$Y_{it} = \alpha \mathrm{SPANK}_i + T_{it}\beta + F_{it} + \mu_{it}, \quad (1)$$

where i indexes the child and t indexes time (T1 or T2). In the model, Y was the outcome of interest (each IDELA developmental domain), SPANK equaled one if the child was spanked the week preceding the survey and zero otherwise, T was a vector of time-varying characteristics (e.g., stimulation and availability of books and toys), F was a vector of time-invariant characteristics (e.g., child sex, time elapse between T1 and T2 and site FE), and μ was residual variation. The coefficient of interest, α, can be interpreted as causal under the conditional independence assumption, which implies that omitted variables like children's genetic makeup or caregivers' mental health are not simultaneously correlated with spanking and children's development after the inclusion of covariates. This may be an implausible assumption but will serve to check how sensitive are findings relative to the within-child analysis described below.

In order to provide more internally valid estimates and assess the robustness of the links between spanking and child developmental outcomes, I conducted a set of child FE models that leveraged within-child variation within the longitudinal data. An individual FE approach allowed me to mitigate issues of confounding by comparing the outcomes of the same child at periods in their life when they are versus are not being spanked, hence controlling for unobserved time-invariant confounders (e.g., genetics, child temperament, country of residence, time elapse between T1 and T2) that are major threats for the validity of between-child comparisons. I estimated the following model:

$$Y_{it} = \alpha \mathrm{SPANK}_i + F_{it} + \theta_i + \mu_{it}, \quad (2)$$

where θ represented a regression intercept for each child. The validity of this model relies on the assumptions that (a) there is variability in spanking within children (Supplemental Table S1 shows variability in spanking between T1 and T2 for all sites) and (b) there are no time-varying confounders left unexplained in the model after the inclusion of covariates. The latter assumption is still strong, but it is more plausible that the assumption from random-effect models as child FE mitigates confounding issues caused by time-fixed confounders. Furthermore, the concurrent use of models that leverage between- and within-child variation provides evidence on the robustness of estimates to different methodological approaches, strengthening the identification of the effect of spanking on children's developmental outcomes.

Table 2 presents details about data missingness. Overall, missingness rates ranged from 0% to 18%. I followed best practice recommendations for handling missing data and employed multiple imputations by chained equations in all models (White et al., 2011), considering that there is no evidence of systematic missingness, imputing 20 data sets based on auxiliary variables with no missing information. As such, all models included the full sample of children. A post hoc power analysis indicated that a total of 3,048 participants was sufficient to detect small effects ($\alpha = .05$, effect size = 0.10, 80% power). Therefore, the present sample size was sufficient to investigate the hypotheses. Furthermore, I standardized all continuous variables to have a mean of 0 and *SD* of 1 to aid the interpretability of the estimates. All analyses were conducted in Stata/MP 17.0.

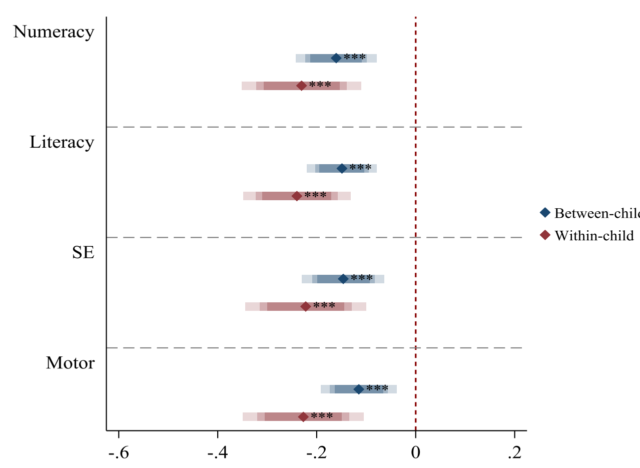
Results

More than 70% of sampled children were spanked in at least one time point (Table 1) and children exhibited developmental progression (i.e., increments in IDELA scores) between T1 and T2 (see Table 2). There were modest but statistically significant negative bivariate associations between spanking and all developmental domains (but early numeracy) ranging from $r = -0.04$.to $r = -0.06$in T1, and more sizable negative correlations in T2, ranging from $r = -0.11$ to $r = -0.21$ (Table 2).

Comparisons between children who were spanked and not spanked (i.e., random-effects models) showed significant links between spanking and worse developmental outcomes in the pooled sample when controlling for children's and their main caregivers' demographic information, other discipline methods, household wealth and learning environments, and binary indicators per site (Figure 1, blue bars, and Table 3, for full results). In particular, children who were

Figure 1

Results From Random-Effects Models Using Between-Child Variation and Fixed-Effects Models Using Within-Child Variation for the Association Between Spanking and Child Developmental Outcomes Using the Pooled Sample



Note. $N = 6{,}096$, corresponding to 3,048 children measured in two time points. Full results are shown in Tables 3 and 4. SE = social–emotional. See the online article for the color version of this figure.
*** $p < .001$.

Table 3
Random-Effects Models on the Association Between Spanking and Developmental Outcomes in the Pooled Sample

Variable	(1) Numeracy	(2) Literacy	(3) Social–emotional	(4) Motor
Spanking	−0.16*** (0.03)	−0.15*** (0.03)	−0.15*** (0.03)	−0.11*** (0.03)
Yelling	0.01 (0.03)	−0.02 (0.02)	0.04 (0.03)	−0.02 (0.03)
Stimulation (*SD*)	0.10*** (0.01)	0.08*** (0.01)	0.12*** (0.01)	0.07*** (0.01)
Child sex (female=1)	−0.01 (0.02)	0.09*** (0.02)	0.04† (0.02)	0.06** (0.02)
Child age in months	0.03*** (0.00)	0.03*** (0.00)	0.02*** (0.00)	0.03*** (0.00)
Mom age: 25–35 years	0.07† (0.04)	0.04 (0.03)	0.12** (0.04)	0.07* (0.04)
Mom age: >36 years	0.09* (0.04)	0.02 (0.04)	0.08† (0.04)	0.05 (0.04)
Mom educ.: primary	−0.03 (0.03)	−0.01 (0.03)	−0.02 (0.04)	−0.02 (0.03)
Mom educ.: secondary	−0.01 (0.04)	−0.02 (0.04)	−0.06 (0.04)	−0.07† (0.04)
Mom educ.: higher	0.14** (0.05)	0.09† (0.05)	−0.02 (0.05)	−0.05 (0.06)
Mom educ.: nonformal	0.14** (0.04)	0.00 (0.04)	0.12** (0.04)	−0.01 (0.04)
Household wealth	0.02† (0.01)	0.04*** (0.01)	0.02 (0.01)	0.05*** (0.01)
Household books (*SD*)	0.06*** (0.01)	0.08*** (0.01)	0.04** (0.02)	0.06*** (0.01)
Household toys (*SD*)	0.20*** (0.02)	0.19*** (0.01)	0.19*** (0.02)	0.18*** (0.01)
Treatment indicator	0.10*** (0.02)	0.12*** (0.02)	0.05* (0.03)	0.09*** (0.02)
Indicator for Cambodia	0.86*** (0.05)	0.91*** (0.04)	0.80*** (0.05)	0.71*** (0.04)
Indicator for Ethiopia	0.65*** (0.04)	0.34*** (0.03)	0.54*** (0.04)	0.31*** (0.04)
Indicator for Rwanda	0.62*** (0.04)	0.77*** (0.04)	0.70*** (0.04)	0.76*** (0.04)
Observations	6,096	6,096	6,096	6,096
Number of children	3,048	3,048	3,048	3,048

Note. Robust standard errors in parentheses. educ. = education.
† $p < .1$. * $p < .05$. ** $p < .01$. *** $p < .001$.

spanked had lower performance than their peers who were not spanked in emergent numeracy (β = −0.16, *SE* = 0.03, $p < .001$), literacy (β = −0.15, *SE* = 0.03, $p < .001$), social–emotional (β = −0.15, *SE* = 0.03, $p < .001$), and motor (β = −0.11, *SE* = 0.03, $p < .001$) skills.

An alternative approach that used within-child comparisons (i.e., FE models) and controlled for time-varying characteristics (including other discipline and the household learning environment) showed stronger links between spanking and negative developmental outcomes in the pooled sample (Figure 1, red bars, and Table 4, for full results). In these models, spanking predicted lower numeracy (β = −0.23, *SE* = 0.05, $p < .001$), literacy (β = −0.24, *SE* = 0.04, $p < .001$), social–emotional (β = −0.22, *SE* = 0.05, $p < .001$), and motor (β = −0.23, *SE* = 0.05, $p < .001$) skills.

Site-specific FE models showed some evidence for heterogeneity in the point estimates between sites (see Figure 2 and Tables 5–8, for full results). In Bhutan, 80% of children were spanked, and spanking predicted lower social–emotional skills by 0.22 *SD* (*SE* = 0.07, $p < .01$) but no other developmental domains. A total of 64% of sampled children in Cambodia were spanked, and there were statistically significant and sizable associations between spanking and numeracy (β = −0.52, *SE* = 0.09, $p < .001$), literacy (β = −0.50, *SE* = 0.09, $p < .001$), and motor (β = −0.40, *SE* = 0.10, $p < .001$) skills, and more modest, marginally significant links with social–emotional development (β = −0.18, *SE* = 0.10, $p < .001$). In Ethiopia, where 85% of children were spanked, while all point estimates indicated a negative association between spanking and developmental outcomes, estimates were imprecise, and no coefficient reached statistical significance. Finally, in Rwanda, about 59% of children were spanked and spanking predicted lower numeracy (β = −0.26, *SE* = 0.10, $p < .005$), literacy (β = −0.33, *SE* = 0.10, $p < .01$), and motor skills (β = −0.33, *SE* = 0.10, $p < .05$).

Discussion

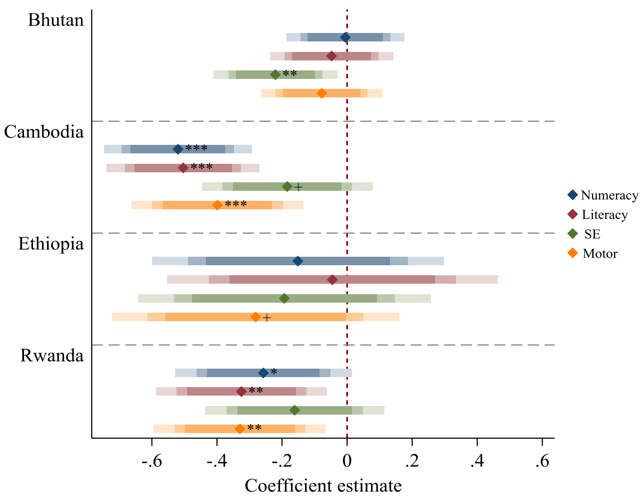
This study combined data from four studies in Bhutan, Cambodia, Ethiopia, and Rwanda regarding preschool-age children's exposure

Table 4
Fixed-Effects Models on the Association Between Spanking and Developmental Outcomes in the Pooled Sample

Variable	(1) Numeracy	(2) Literacy	(3) Social–emotional	(4) Motor
Spanking	−0.23*** (0.05)	−0.24*** (0.04)	−0.22*** (0.05)	−0.23*** (0.05)
Yelling	0.02 (0.04)	0.01 (0.04)	0.03 (0.04)	−0.09* (0.04)
Stimulation (*SD*)	0.16*** (0.02)	0.15*** (0.02)	0.19*** (0.02)	0.13*** (0.02)
Household books (*SD*)	0.13*** (0.03)	0.12*** (0.02)	0.08** (0.03)	0.10*** (0.02)
Household toys (*SD*)	0.33*** (0.02)	0.31*** (0.02)	0.31*** (0.02)	0.30*** (0.02)
Observations	6,096	6,096	6,096	6,096
Number of children	3,048	3,048	3,048	3,048

Note. Robust standard errors in parentheses.
* $p < .05$. ** $p < .01$. *** $p < .001$.

Figure 2
Results From Fixed-Effects Models Using Within-Child Variation for the Association Between Spanking and Child Developmental Outcomes by Country



Note. $N = 6{,}096$, corresponding to 3,048 children measured in two time points. Full results are shown in Tables 5–8. SE = social–emotional. See the online article for the color version of this figure.
$^{\dagger} p < .1$. $^{**} p < .01$. $^{***} p < .001$.

to spanking and direct assessments of their numeracy, literacy, social–emotional, and motor skills. I leveraged the longitudinal design of the studies to use both between- and within-child variation in exposure to spanking, providing more internally valid evidence on the potential developmental consequences of spanking and examining consistency in associations across settings. The results from random-effects and more statistically conservative fixed-effects models using the pooled sample indicate that spanking predicted lower developmental skills, with effect sizes ranging from −0.11 to −0.24 *SD*. These point estimates are smaller than the average effect size ($d = -0.33$, 95%CI [−0.29, −0.38]). identified in a meta-analysis of 111 bivariate effect sizes representing 160,927 children, mostly from the United States (Gershoff & Grogan-Kaylor, 2016), but are similar to the effect sizes found in studies using longitudinal samples and stronger methodological designs (e.g., $d_{\text{range}} = -0.04, -0.21$; Cuartas et al., 2020; Gershoff et al., 2018; Kang, 2022; Okuzono et al., 2017). Somewhat unexpectedly, estimates for the FE models were larger than those from random-effect models, which warrants further research.

The use of direct assessments of children's development with IDELA allowed me to examine the associations between spanking and different developmental domains with higher precision relative to prior cross-cultural studies that relied on basic caregiver reports of children's skills and behaviors (e.g., Cuartas, 2021; Grogan-Kaylor et al., 2021; Pace et al., 2019). In line with multiple theories of human development and my a priori hypothesis, the current findings indicate that spanking has the potential to undermine early cognitive and social–emotional development in ways that threaten lifelong developmental trajectories. Indeed, prior studies have already identified associations between spanking and negative cognitive, regulatory, and emotional outcomes that are closely related to the onset of early internalizing and externalizing behavior problems, learning challenges, and future health difficulties (Cuartas, 2022; Gershoff & Grogan-Kaylor, 2016; Heilmann et al., 2021). Yet, spanking is associated with negative motor outcomes in the current sample, contradicting my stated hypothesis. One possibility is that by employing spanking, parents restrict children's movement behaviors and physical exploration, impacting their motor development. This hypothesis remains to be tested.

The site-specific estimates show that spanking is not related to any positive developmental outcome in any site. On the contrary, spanking predicted multiple negative developmental outcomes, but there was sizable heterogeneity between sites. While all point estimates were negative, many were not statistically significant. Using the present data, I cannot definitively explain why this variability is occurring. However, the lack of statistically significant links in the sample from Ethiopia might be due to little variation in spanking, as most of the sampled children were spanked at both time points. For the rest of the sites, several factors might contribute to the observed variation, including differences in children's age at T1 (see Table 1), the gap between T1 and T2, and measurement error caused by cultural differences. Indeed, ethnographic evidence has shown variability in daily interactions, the ways the same behaviors manifest across cultures, and the way caregivers understand the same concept (Cheung et al., 2021; Gaskins, 2015). Subtle differences like the way spanking is administered (i.e., frequency or severity) or interpreted by parents or children in different settings, which I cannot observe in the current data, might

Table 5
Fixed-Effects Models on the Association Between Spanking and Developmental Outcomes in Bhutan

Variable	(1) Numeracy	(2) Literacy	(3) Social–emotional	(4) Motor
Spanking	−0.01 (0.07)	−0.05 (0.07)	-0.22^{**} (0.07)	−0.08 (0.07)
Yelling	−0.03 (0.06)	−0.04 (0.06)	0.04 (0.06)	-0.16^{**} (0.06)
Stimulation (*SD*)	0.03^{*} (0.01)	0.05^{**} (0.01)	0.07^{***} (0.01)	0.05^{***} (0.01)
Household books (*SD*)	0.07^{**} (0.02)	0.07^{**} (0.02)	$0.04^{\dagger}$ (0.03)	0.07^{**} (0.02)
Household toys (*SD*)	0.15^{***} (0.02)	0.16^{***} (0.02)	0.16^{***} (0.02)	0.17^{***} (0.02)
Observations	2,754	2,754	2,754	2,754
Number of children	1,377	1,377	1,377	1,377

Note. Robust standard errors in parentheses.
$^{\dagger} p < .1$. $^{*} p < .05$. $^{**} p < .01$. $^{***} p < .001$.

Table 6
Fixed-Effects Models on the Association Between Spanking and Developmental Outcomes in Cambodia

Variable	(1) Numeracy	(2) Literacy	(3) Social–emotional	(4) Motor
Spanking	-0.52^{***} (0.09)	-0.50^{***} (0.09)	$-0.18^{\dagger}$ (0.10)	-0.40^{***} (0.10)
Yelling	-0.33^{**} (0.11)	-0.32^{**} (0.11)	-0.33^{**} (0.12)	-0.33^{**} (0.12)
Stimulation (*SD*)	−0.04 (0.03)	−0.02 (0.03)	−0.04 (0.03)	−0.00 (0.03)
Household books (*SD*)	0.18^{**} (0.06)	0.20^{***} (0.05)	$0.10^{\dagger}$ (0.06)	0.16^{**} (0.06)
Household toys (*SD*)	0.25^{***} (0.02)	0.24^{***} (0.02)	0.20^{***} (0.03)	0.21^{***} (0.02)
Observations	764	764	764	764
Number of children	382	382	382	382

Note. Robust standard errors in parentheses.
$^{\dagger}p < .1$. $^{**}p < .01$. $^{***}p < .001$.

be relevant to explain the heterogeneity and suggest the need for further culturally nuanced research in LMICs.

Limitations

The four data sets included in this study measured spanking using a single item on whether the behavior occurred in the previous week, which has become standard in the study of physical punishment (Gershoff & Grogan-Kaylor, 2016). While such a measurement approach seeks to reduce recall bias and address the impossibility of measuring spanking by direct observation (as it is a low-incidence behavior), it might misestimate the actual use of spanking if such behavior varies in how much it occurs before the 1-week period. Yet, misclassifying "spanked" children as "not spanked" might lead to an underestimation of the "true" consequences of spanking. Nonetheless, the use of a single item makes it difficult to assess variation in the consequences of spanking given its frequency, severity, and how normative, acceptable, or justifiable is the behavior for caregivers and children.

Moreover, the data sets used in this study did not include other important time-varying confounders that could threaten the internal validity of estimates. For example, spanking is often associated with other forms of family violence like physical and psychological abuse and intimate partner violence (Cerna-Turoff et al., 2021), which could contribute to the developmental outcomes assessed but were not included as covariates in the present study. Furthermore, the current research employs information from studies conducted in four countries with two rounds of data collection, which are not nationally representative, so the results have limited external validity and can only speak to short-term "effects" of spanking. Even though the goal of this study was to assess the links between spanking and child developmental outcomes in underrepresented contexts, there are limited comparable longitudinal data sets with detailed measures of development and multiple rounds of data collection.

Future Research Directions

Future research should complement the current body of evidence by employing more comprehensive measurement instruments that allow researchers to assess how specific characteristics of spanking, like its frequency, severity, or how normative it is, might lead to variation in its associations with developmental outcomes in LMICs. Similarly, future research should collect more detailed information on parenting, other forms of discipline and punishment, and violence to strengthen the set of time-varying covariates, mitigate threats to the FE models, and assess differential associations between spanking and child outcomes due to contextual characteristics (e.g., parental warmth and sensitiveness). Finally, future studies would benefit from using cohort studies and other longitudinal data with more than two points of data collection to examine the both short- and long-term developmental consequences of experiencing spanking during early childhood.

Prevention and Policy Implications

The present study's detailed and more conservative set of estimates complements a vast evidence base from the United States and other high-income countries and nascent evidence from LMICs on the potential negative developmental consequences of spanking

Table 7
Fixed-Effects Models on the Association Between Spanking and Developmental Outcomes in Ethiopia

Variable	(1) Numeracy	(2) Literacy	(3) Social–emotional	(4) Motor
Spanking	−0.15 (0.17)	−0.05 (0.19)	−0.19 (0.17)	$-0.28^{\dagger}$ (0.17)
Yelling	0.17 (0.14)	$0.29^{\dagger}$ (0.16)	0.08 (0.16)	0.35^{*} (0.14)
Stimulation (*SD*)	0.12^{***} (0.02)	0.09^{***} (0.02)	0.13^{***} (0.02)	0.08^{***} (0.02)
Household books (*SD*)	0.04 (0.04)	0.03 (0.03)	0.01 (0.03)	0.01 (0.03)
Household toys (*SD*)	0.06^{**} (0.02)	0.07^{***} (0.02)	0.07^{**} (0.02)	0.05^{**} (0.02)
Observations	1,386	1,386	1,386	1,386
Number of children	693	693	693	693

Note. Robust standard errors in parentheses.
$^{\dagger}p < .1$. $^{**}p < .01$. $^{***}p < .001$.

Table 8
Fixed-Effects Models on the Association Between Spanking and Developmental Outcomes in Rwanda

Variable	(1) Numeracy	(2) Literacy	(3) Social–emotional	(4) Motor
Spanking	−0.26* (0.10)	−0.33** (0.10)	−0.16 (0.11)	−0.33** (0.10)
Yelling	0.04 (0.10)	0.05 (0.10)	−0.00 (0.10)	0.00 (0.10)
Stimulation (*SD*)	0.08*** (0.02)	0.08*** (0.02)	0.07*** (0.02)	0.06*** (0.02)
Household books (*SD*)	0.18*** (0.05)	0.19*** (0.05)	0.10* (0.05)	0.15** (0.05)
Household toys (*SD*)	0.16*** (0.03)	0.18*** (0.03)	0.16*** (0.03)	0.17*** (0.03)
Observations	1,192	1,192	1,192	1,192
Number of children	596	596	596	596

Note. Robust standard errors in parentheses.
* $p < .05$. ** $p < .01$. *** $p < .001$.

(Cuartas, 2023; Gershoff & Grogan-Kaylor, 2016; Heilmann et al., 2021). To date, there is no evidence that spanking might be beneficial for children's cognitive, behavioral, or social–emotional development, and existing evidence overwhelmingly indicates that spanking compromises children's development. The existing evidence and basic considerations of children's rights (United Nations Committee on the Rights of the Child, 2007) indicate that policymakers and practitioners should work on the prevention and reduction of spanking and all other forms of physical punishment to promote children's positive developmental trajectories. Some promising approaches to doing so are legislation to protect children's right to safety and security and, most importantly, parenting programs, which can promote behavior change, positive parenting, and the prevention of violence in LMICs (Coore Desai et al., 2017).

References

Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Prentice Hall.

Baumrind, D. (1997). Necessary distinctions. *Psychological Inquiry*, *8*(3), 176–182. https://doi.org/10.1207/s15327965pli0803_2

Berens, A. E., & Nelson, C. A. (2019). Neurobiology of fetal and infant development: Implications for infant mental health. In C. H. Zeanah (Ed.), *Handbook of infant mental health* (pp. 41–62). Guilford Press.

Berry, D., & Willoughby, M. T. (2017). On the practical interpretability of cross-lagged panel models: Rethinking a developmental workhorse. *Child Development*, *88*(4), 1186–1206. https://doi.org/10.1111/cdev.12660

Bhatia, A., Fabbri, C., Cerna-Turoff, I., Turner, E., Lokot, M., Warria, A., Tuladhar, S., Tanton, C., Knight, L., Lees, S., Cislaghi, B., Bhabha, J., Peterman, A., Guedes, A., & Devries, K. (2021). Violence against children during the COVID-19 pandemic. *Bulletin of the World Health Organization*, *99*(10), 730–738. https://doi.org/10.2471/BLT.20.283051

Black, M. M., Walker, S. P., Fernald, L. C. H., Andersen, C. T., DiGirolamo, A. M., Lu, C., McCoy, D. C., Fink, G., Shawar, Y. R., Shiffman, J., Devercelli, A. E., Wodon, Q. T., Vargas-Barón, E., Grantham-McGregor, S., & the Lancet Early Childhood Development Series Steering Committee. (2017). Early childhood development coming of age: Science through the life course. *The Lancet*, *389*(10064), 77–90. https://doi.org/10.1016/S0140-6736(16)31389-7

Bowlby, J. (1969). *Attachment and loss*. Basic Books.

Bugental, D. B., Martorell, G. A., & Barraza, V. (2003). The hormonal costs of subtle forms of infant maltreatment. *Hormones and Behavior*, *43*(1), 237–244. https://doi.org/10.1016/S0018-506X(02)00008-9

Cerna-Turoff, I., Fang, Z., Meierkord, A., Wu, Z., Yanguela, J., Bangirana, C. A., & Meinck, F. (2021). Factors associated with violence against children in low- and middle-income countries: A systematic review and meta-regression of nationally representative data. *Trauma, Violence, & Abuse*, *22*(2), 219–232. https://doi.org/10.1177/1524838020985532

Cheung, S. K., Dulay, K. M., Yang, X., Mohseni, F., & McBride, C. (2021). Home literacy and numeracy environments in Asia. *Frontiers in Psychology*, *12*, Article 578764. https://doi.org/10.3389/fpsyg.2021.578764

Coore Desai, C., Reece, J.-A., & Shakespeare-Pellington, S. (2017). The prevention of violence in childhood through parenting programmes: A global review. *Psychology, Health & Medicine*, *22*(Suppl. 1), 166–186. https://doi.org/10.1080/13548506.2016.1271952

Cuartas, J. (2021). Corporal punishment and early childhood development in 49 low- and middle-income countries. *Child Abuse & Neglect*, *120*, Article 105205. https://doi.org/10.1016/j.chiabu.2021.105205

Cuartas, J. (2022). The effect of spanking on early social–emotional skills. *Child Development*, *93*(1), 180–193. https://doi.org/10.1111/cdev.13646

Cuartas, J. (2023). Corporal punishment and child development in low- and-middle-income countries: Progress, challenges, and directions. *Child Psychiatry and Human Development*. Advance online publication. https://doi.org/10.1007/s10578-022-01362-3

Cuartas, J., McCoy, D., Sánchez, J., Behrman, J., Cappa, C., Donati, G., Heymann, J., Lu, C., Raikes, A., Rao, N., Richter, L., Stein, A., & Yoshikawa, H. (2023). Family play, reading, and other stimulation and early childhood development in five low-and-middle-income countries. *Developmental Science*, *26*(6), Article e13404. https://doi.org/10.1111/desc.13404

Cuartas, J., McCoy, D. C., Grogan-Kaylor, A., & Gershoff, E. (2020). Physical punishment as a predictor of early cognitive development: Evidence from econometric approaches. *Developmental Psychology*, *56*(11), 2013–2026. https://doi.org/10.1037/dev0001114

Cuartas, J., Weissman, D. G., Sheridan, M. A., Lengua, L., & McLaughlin, K. A. (2021). Corporal punishment and elevated neural response to threat in children. *Child Development*, *92*(3), 821–832. https://doi.org/10.1111/cdev.13565

Deater-Deckard, K., & Dodge, K. A. (1997). Externalizing behavior problems and discipline revisited: Nonlinear effects and variation by culture, context, and gender. *Psychological Inquiry*, *8*(3), 161–175. https://doi.org/10.1207/s15327965pli0803_1

Dobbs, T., & Duncan, J. (2004). Children's perspectives on physical discipline: A New Zealand example. *Child Care in Practice*, *10*(4), 367–379. https://doi.org/10.1080/1357527042000285547

Dodge, K. A. (1993). Social-cognitive mechanisms in the development of conduct disorder and depression. *Annual Review of Psychology*, *44*(1), 559–584. https://doi.org/10.1146/annurev.ps.44.020193.003015

Draper, C. E., Barnett, L. M., Cook, C. J., Cuartas, J. A., Howard, S. J., McCoy, D. C., Merkley, R., Molano, A., Maldonado-Carreño, C., Obradović, J., Scerif, G., Valentini, N. C., Venetsanou, F., & Yousafzai, A. K. (2022). Publishing child development research from around the world: An unfair playing field resulting in most of the world's child population under-represented in research. *Infant and Child Development*, *32*(6), Article e2375. https://doi.org/10.1002/icd.2375

Duncan, G. J., Magnuson, K. A., & Ludwig, J. (2004). The endogeneity problem in developmental studies. *Research in Human Development*, *1*(1–2), 59–80. https://doi.org/10.1080/15427609.2004.9683330

Gaskins, S. (2015). Childhood practices across cultures: Play and household work. In L. A. Jensen (Ed.), *The Oxford handbook of human development and culture: An interdisciplinary perspective* (pp. 185–197). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199948550.001.0001

Gershoff, E. T. (2002). Corporal punishment by parents and associated child behaviors and experiences: A meta-analytic and theoretical review. *Psychological Bulletin*, *128*(4), 539–579. https://doi.org/10.1037/0033-2909.128.4.539

Gershoff, E. T., & Grogan-Kaylor, A. (2016). Spanking and child outcomes: Old controversies and new meta-analyses. *Journal of Family Psychology*, *30*(4), 453–469. https://doi.org/10.1037/fam0000191

Gershoff, E. T., Grogan-Kaylor, A., Lansford, J. E., Chang, L., Zelli, A., Deater-Deckard, K., & Dodge, K. A. (2010). Parent discipline practices in an international sample: Associations with child behaviors and moderation by perceived normativeness. *Child Development*, *81*(2), 487–502. https://doi.org/10.1111/j.1467-8624.2009.01409.x

Gershoff, E. T., Sattler, K. M. P., & Ansari, A. (2018). Strengthening causal estimates for links between spanking and children's externalizing behavior problems. *Psychological Science*, *29*(1), 110–120. https://doi.org/10.1177/0956797617729816

Gonzáles, M. R., Trujillo, A., & Carvalho, G. (2019). *Castigo físico en Colombia la voz de los niños, las niñas y los adolescentes: aportes para una educación fundamentada en derechos humanos*. Educrítica.

Grogan-Kaylor, A., Castillo, B., Pace, G. T., Ward, K. P., Ma, J., Lee, S. J., & Knauer, H. (2021). Global perspectives on physical and nonphysical discipline: A Bayesian multilevel analysis. *International Journal of Behavioral Development*, *45*(3), 216–225. https://doi.org/10.1177/0165025420981642

Halpin, P. F., Wolf, S., Yoshikawa, H., Rojas, N., Kabay, S., Pisani, L., & Dowd, A. J. (2019). Measuring early learning and development across cultures: Invariance of the IDELA across five countries. *Developmental Psychology*, *55*(1), 23–37. https://doi.org/10.1037/dev0000626

Heilmann, A., Mehay, A., Watt, R. G., Kelly, Y., Durrant, J. E., van Turnhout, J., & Gershoff, E. T. (2021). Physical punishment and child outcomes: A narrative review of prospective studies. *The Lancet*, *398*(10297), 355–364. https://doi.org/10.1016/S0140-6736(21)00582-1

Iwamoto, S., Abimpaye, M., & Mukantagwera, L. (2019). *Advancing the school readiness 4–6 program in Rwanda endline report*. https://idela-network.org/wp-content/uploads/2020/08/Ferrari-ASR-Impact-Evaluation-Report_2019.pdf

Kang, J. (2022). Spanking and children's early academic skills: Strengthening causal estimates. *Early Childhood Research Quarterly*, *61*, 47–57. https://doi.org/10.1016/j.ecresq.2022.05.005

Larzelere, R. E., Gunnoe, M. L., & Ferguson, C. J. (2018). Improving causal inferences in meta-analyses of longitudinal studies: Spanking as an illustration. *Child Development*, *89*(6), 2038–2050. https://doi.org/10.1111/cdev.13097

McCoy, D. C., Seiden, J., Cuartas, J., Pisani, L., & Waldman, M. (2022). Estimates of a multidimensional index of nurturing care in the next 1000 days of life for children in low-income and middle-income countries: A modelling study. *The Lancet: Child & Adolescent Health*, *6*(5), 324–334. https://doi.org/10.1016/S2352-4642(22)00076-1

Mesman, J., Minter, T., Angnged, A., Cissé, I. A. H., Salali, G. D., & Migliano, A. B. (2018). Universality without uniformity: A culturally inclusive approach to sensitive responsiveness in infant caregiving. *Child Development*, *89*(3), 837–850. https://doi.org/10.1111/cdev.12795

Okuzono, S., Fujiwara, T., Kato, T., & Kawachi, I. (2017). Spanking and subsequent behavioral problems in toddlers: A propensity score-matched, prospective study in Japan. *Child Abuse & Neglect*, *69*, 62–71. https://doi.org/10.1016/j.chiabu.2017.04.002

Pace, G. T., Lee, S. J., & Grogan-Kaylor, A. (2019). Spanking and young children's socioemotional development in low- and middle-income countries. *Child Abuse & Neglect*, *88*, 84–95. https://doi.org/10.1016/j.chiabu.2018.11.003

Pisani, L., Borisova, I., & Dowd, A. J. (2018). Developing and validating the International Development and Early Learning Assessment (IDELA). *International Journal of Educational Research*, *91*, 1–15. https://doi.org/10.1016/j.ijer.2018.06.007

Pisani, L., Dib, G., & Khoy, R. (2016). *Cambodia first read endline report*. https://resourcecentre.savethechildren.net/document/cambodia-first-read-endline-report-2016/

Pisani, L., Dyenka, K., Sharma, P., Chhetri, N., Dang, S., Gayleg, K., & Wangdi, C. (2017). Bhutan's national ECCD impact evaluation: Local, national, and global perspectives. *Early Child Development and Care*, *187*(10), 1511–1527. https://doi.org/10.1080/03004430.2017.1302944

Save the Children. (2015). *National ECCD impact evaluation study 2015*. https://resourcecentre.savethechildren.net/library/national-eccd-impact-evaluation-study-2015

Seiden, J., Yenew, A., Kefey, K., Abrha, H., & Marino, J. (2018). *Central Tigray sponsorship longitudinal study of learning and development: Baseline results*. Save the Children.

United Nations Committee on the Rights of the Child. (2007). *CRC General Comment No. 8 (2006): The right of the child to protection from corporal punishment and other cruel or degrading forms of punishment* (U.N. CRC/C/GC/8).

Waldman, M., McCoy, D. C., Seiden, J., Cuartas, J., & Fink, G. (2021). Validation of motor, cognitive, language, and socio-emotional subscales using the caregiver reported early development instruments: An application of multidimensional item factor analysis. *International Journal of Behavioral Development*, *45*(4), 368–377. https://doi.org/10.1177/01650254211005560

White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, *30*(4), 377–399. https://doi.org/10.1002/sim.4067

Wolf, S., Halpin, P., Yoshikawa, H., Dowd, A. J., Pisani, L., & Borisova, I. (2017). Measuring school readiness globally: Assessing the construct validity and measurement invariance of the International Development and Early Learning Assessment (IDELA) in Ethiopia. *Early Childhood Research Quarterly*, *41*, 21–36. https://doi.org/10.1016/j.ecresq.2017.05.001

Received November 8, 2022
Revision received November 6, 2023
Accepted November 8, 2023 ■